INSTRUMENTATION · HOW-TO GUIDE

Extending a Type K Thermocouple Without Corrupting the Reading

A step-by-step guide to safely joining a 5 metre extension wire to a Type K thermocouple probe — written for someone doing this for the first time.

Sensor type: Type K **Goal:** reach 1 m → 5 m **Reads into:** MAX6675 module

Skill needed: beginner

✓ **Yes — these two parts are compatible.**

Both the probe and the extension wire are **Type K**. That is the one thing that must match, and it does. You can join them — you just have to follow the rules below, because a careless join is the number-one cause of noisy or wrong temperature readings.

01 · WHAT YOU HAVE

The two parts

THE SENSOR

MAX6675 module + short K probe

Thermocouple type	K ✓
Probe length	~0.45 m
Reads temperature	0–1024 °C
Bundled probe rated	~80 °C

THE EXTENSION

RS Pro Type K wire, 5 m

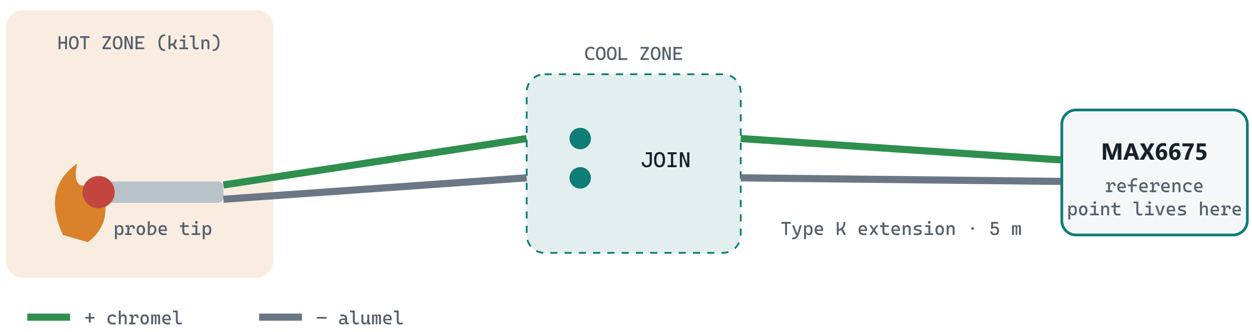
Thermocouple type	K ✓
Length	5 m
Cores	2 (flat pair)
Rated to	350 °C

Because **both say "Type K"**, they speak the same electrical language. If one had been Type J, T or another letter, they could **not** be joined — mixing types creates an error that can never be corrected.

02 · THE IDEA IN ONE PICTURE

Why the extension wire must also be Type K

A thermocouple works by comparing the *hot tip* against a *reference point*. On this setup the reference point lives **on the MAX6675 board** — that is where it measures "room temperature" to do its maths. For that to be correct, Type K wire has to run **all the way** from the tip to the board. The extension is simply more Type K wire, so the reference point stays exactly where it belongs.



Type K runs unbroken from tip → join → module. Keep the join in the cool zone.

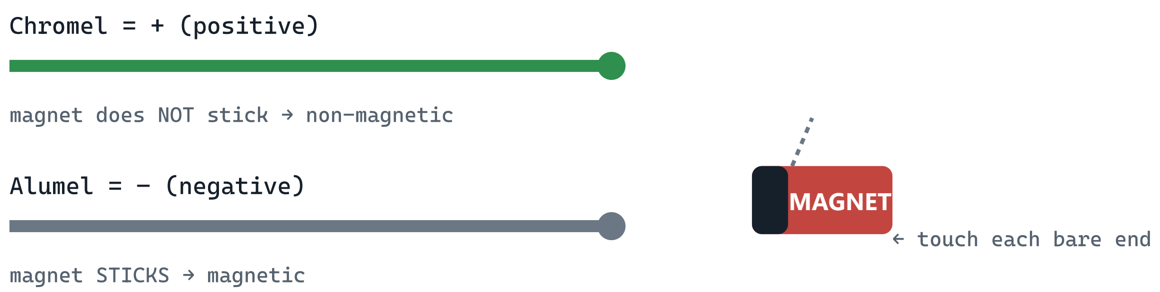
> **Why this matters:** if you extended with ordinary **copper** wire instead, the reference point would jump to the join and your readings would drift with the temperature of that spot. Always extend with the Type K wire only.

03 · BEFORE YOU JOIN – FIND THE + AND –

Every Type K wire has a positive and a negative side

A Type K wire is made of two different metals: **chromel** (the positive, +) and **alumel** (the negative, –). You must connect **+ to + and – to –**. If you cross them, the reading will run backwards or be badly wrong.

Colours are a hint but vary by country, so **confirm with a magnet** — this always works:



The wire the magnet grabs is the negative (–). Do this on both the probe and the extension.

> **Colour cross-check:** UK / European (IEC) wire uses **green = +** and **white = –**. US (ANSI) wire uses **yellow = +** and **red = –**. When in doubt, the magnet wins.

04 · WHAT TO GATHER

Tools & materials

ALWAYS NEEDED

Basics

Wire strippers	✓
Small magnet	✓
Heat-shrink / tape	✓

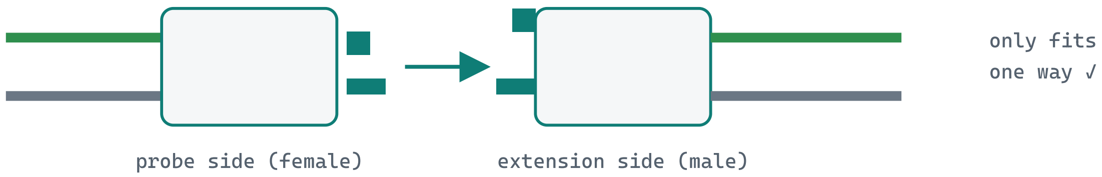
PICK ONE METHOD

To make the join

K connector pair	best
— or —	
Crimp ferrules	ok

RECOMMENDED Method A — Type K connectors

A miniature Type K connector pair (a flat-pin male and female, usually green or yellow) is the safest option. It is **keyed**, so it can only plug in one way — you physically cannot reverse the polarity — and it is made of K alloy, so it adds no error.



The different-sized pins mean the connector cannot go in backwards.

1 Fit a connector to each end

Attach the **female** connector to the probe leads and the **male** connector to one end of the 5 m wire. Each connector has a marked + and – terminal — match them using your magnet test.

2 Plug them together

Push the two halves together. They mate in only one orientation, so polarity is automatically correct.

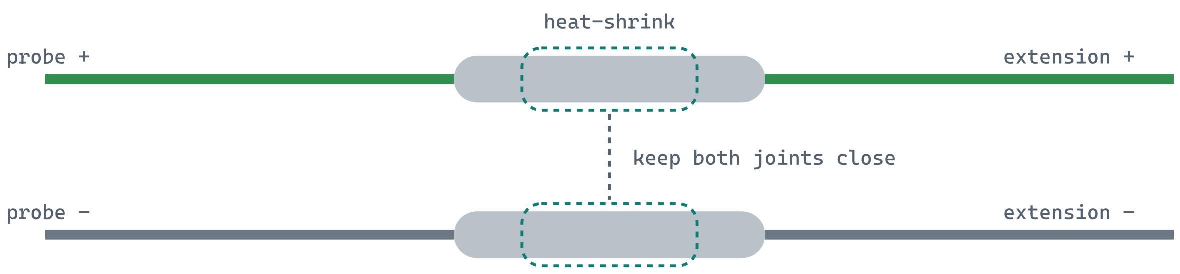
3 Wire the far end into the MAX6675

Connect the other end of the 5 m wire to the module's screw terminals — **+ wire to the + terminal, – wire to the – terminal.**

NO CONNECTORS? DO THIS

Method B — crimp splice

If you have no connectors, a clean crimped splice is reliable. The key rules: keep the two joints **separate and insulated** so they can never touch, and keep them **side by side** so they stay at the same temperature.



Two insulated crimps, side by side. The + and − must never touch each other.

1 Strip the ends

Strip about **8–10 mm** of insulation off all four wire ends (two on the probe, two on the extension).

2 Sort by polarity

Use the magnet to identify + and – on each. Lay out the two **+** wires together and the two **–** wires together.

3 Crimp each pair separately

Twist **+ to +** and slide into one insulated crimp ferrule; twist **– to –** into a second ferrule. Squeeze each with the crimp tool. Crimping beats bare twisting — twists loosen and corrode, which makes the reading jump around.

4 Insulate both joints

Cover each joint with its own heat-shrink so the + and – can never touch. Keep the two joints next to each other.

5

Wire into the MAX6675

Land the far end on the module's screw terminals — + to +, – to –.

06 · AVOID THESE MISTAKES

The four things that cause noise & wrong readings

- **Don't use copper wire** to extend. It moves the reference point and makes readings drift. Only Type K wire.
- **Don't reverse polarity.** Crossing + and – makes the temperature read backwards or far off. Always magnet-test.
- **Don't solder the join.** Solder adds a third metal and can throw off the reading; a crimp or connector is safer for a beginner.
- **Don't put the join in the heat.** Keep the splice in a cool spot, away from the kiln — the extension wire is only rated to 350 °C.

07 · CHECK IT WORKS

Three quick tests after joining

1 **Room-temperature check.** With the probe just sitting in the room, the reading should be close to the actual room temperature (~25–30 °C).

2 **Warm the tip.** Hold the probe tip in your fingers — the number should **rise** a few degrees. If it **drops**, your polarity is reversed → swap the + and – at the join.

3 **Wiggle test.** Gently move the join while watching the number. If it jumps around, the connection is loose → re-crimp or re-seat the connector.

08 · IMPORTANT FOR THE KILN

The bundled probe is only rated to ~80 °C

A Kon-Tiki kiln runs at 400–700 °C, which would destroy the little probe that comes with the MAX6675 almost instantly. The MAX6675 chip itself can read up to ~1024 °C, but it needs a proper **high-temperature Type K probe** (mineral-insulated, with a stainless or ceramic sheath, rated ≥ 800 °C).

Extend **that** high-temperature probe using the exact same method in this guide. The extension wire's 350 °C rating is fine, because only the probe tip goes in the fire — the join and the wire stay in the cool zone.

Prepared as a field reference for joining a Type K thermocouple to a Type K extension wire · Both parts verified Type K & compatible.